

Exact Operator Structure and Finite Information Memory in Accelerated Collatz Dynamics

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Abstract

We study the Collatz first-return map on Type A integers $\mathcal{A} = \{n \equiv 3 \pmod{4}\}$ via a 2-adic symbolic encoding. The main results are:

- (Theorem EC)** The coding map Φ is a homeomorphism from the 2-adic completion $\hat{\mathcal{A}} \subset \mathbb{Z}_2$ onto a countable-state renewal shift (X, σ) whose stationary distribution is Geometric $(1/2)$; positive integers inject as a dense arithmetic subset.
- (Lemma S1 — Shift-and-Kill)** The Markov operator satisfies $P^N f = S^N f$ exactly on mean-zero functions, where $(Sf)(j) = f(j-1) \cdot \mathbf{1}_{\{j \geq 2\}}$. This single identity, proved by one-step induction, is the structural core.
- (Theorem 25-I)** From Lemma S1: the exact covariance $\mathrm{Cov}(\pi(k_0), \pi(k_N)) = 2 \cdot (1/2)^N$, the L^∞ mixing bound $\|\mathrm{Cov}\| \leq \|f\| \cdot \|g\| \cdot (1/2)^N$, and the L^2 decay $\|(1/\sqrt{2})^N$.
- (Theorem IT2a)** The total stationary mutual information is $\sum_{N=1}^\infty I(\pi(k_0; k_N)) = 2 \log 2$ (a finite, explicitly computable value; numerically equal to the measure-theoretic entropy of one renewal step).

All results are rigorous. The paper does not address Collatz convergence; it establishes the exact operator structure and spectral properties of the induced renewal system.

Keywords: Collatz map, renewal shift, 2-adic completion, Markov operator, shift-and-kill, mutual information, ergodic theory.

1. Introduction

1.1 The Renewal Structure of Collatz

The Collatz map $T(n) = n/2$ (even) or $(3n+1)/2^{v_2(3n+1)}$ (odd) admits a symbolic encoding via the first-return map to Type A integers $\mathcal{A} = \{n \equiv 3 \pmod{4}\}$. The key observation: the integer $k(n) = v_2(n+1) - 1$ attached to each Type A point evolves under $T|_{\mathcal{A}}$ as a **renewal chain** — state k descends deterministically to $k-1$ (for $k \geq 2$) or renews to a fresh Geometric $(1/2)$ draw (for $k=1$). This coding extends to an exact 2-adic homeomorphism on the completion $\hat{\mathcal{A}}$, with positive integers forming a dense arithmetic subset.

1.2 The Shift-and-Kill Decomposition

The central contribution is **Lemma S1**:

$$S^N f = S^N f \quad \text{for all } f \text{ with } E_{\pi}[f] = 0,$$

where S is the shift-and-kill operator. The proof is a one-step induction on N : the renewal state sends mean-zero functions to zero ($(Pf)(1) = E_{\pi}[f] = 0 = (Sf)(1)$), so S^N and S agree on $\ker E_{\pi}$ for one step, and the identity propagates by induction.

From this single identity, all mixing and memory results follow immediately.

1.3 Scope

This paper establishes the exact operator structure of the induced symbolic renewal system. It does **not** prove the Collatz conjecture, address whether divergent orbits exist, or connect the renewal structure to arithmetic number theory. These questions are the subject of the companion paper (Kamanoi 2026b).

2. Type A Coding

2.1 Definitions

$T(n) = n/2$ (even) or $(3n+1)/2^{v_2(3n+1)}$ (odd). $\mathcal{A} = \{n \in \mathbb{Z}_{>0} : n \equiv 3 \pmod{4}\}$. $\hat{\mathcal{A}} = \{x \in \mathbb{Z}_2 : x \equiv 3 \pmod{4}\}$. For $n \in \mathcal{A}$: $k(n) = v_2(n+1) - 1 \geq 1$ (well-defined since $v_2(n+1) \geq 2$). $T|_{\mathcal{A}}(n) =$ first return to $\mathcal{A}$ under T .

Renewal shift (X, σ) : the topological shift on alphabet $\{1, 2, \dots\}$ endowed with the Markov transition kernel $P(j, j-1) = 1$ ($j \geq 2$) and $P(1, j) = 1/2^j$ ($j \geq 1$).

2.2 Lemma E [R]

Lemma E. For $K \geq 2$ and $a \equiv 3 \pmod{4}$: $\frac{\#\{n \leq N: n \in \mathcal{A}, n \equiv a \pmod{2^K}\}}{\#\{n \leq N: n \in \mathcal{A}\}} = \frac{1}{2^{K-2}} + O(1/N)$.

Proof. Numerator $= N/2^K + O(1)$; denominator $= N/4 + O(1)$; ratio $= 4/2^K = 1/2^{K-2}$. $\square$

2.3 Lemma A [R]

Lemma A. For $n \in \mathcal{A}$ with $k(n)=j \geq 2$: $T(n) \in \mathcal{A}$ and $k(T(n))=j-1$.

Proof. $n=2^{j+1}q-1$ (odd q); $3n+1=2(3 \cdot 2^{jq}-1)$ with $v_2=1$ (since $j \geq 2$); $T(n)=3 \cdot 2^{jq}-1 \equiv 3 \pmod{4}$; $k(T(n))=j-1$. $\square$

[625 cases, $n \leq 5000$, 0 errors.]

2.4 Lemma B [R]

Lemma B (under Lemma E). For $k(n)=1$, n uniform from $\mathcal{A} \cap [1, N]$ as $N \rightarrow \infty$: $P(k(T(\mathcal{A}(n)))=j) \rightarrow \frac{1}{2^j} \quad \text{as } N \rightarrow \infty$.

Proof. For $n=8m+3$: $T^2(n)=(9m+4)/2^{v_2(9m+4)}$. Since $\gcd(9, 2^K)=1$, the map $m \mapsto 9m+4$ is a bijection mod 2^K ; Lemma E then gives equidistribution of $9m+4 \pmod{2^K}$, hence $v_2(9m+4) \rightarrow \text{Geom}(1/2)$ in distribution and the odd part equidistributed. It follows that $P(T^2(n) \in \mathcal{A}) \rightarrow 1/2$ and, conditional on $T^2(n) \in \mathcal{A}$: $k(T^2(n)) \rightarrow \text{Geom}(1/2)$ in distribution. $\square$

[24,861 samples, all $|z| < 1.5$.]

3. The 2-adic Symbolic Model

3.1 Theorem EC [R]

Theorem EC.

(a) Injection. $\Phi(n)=(k(n), k(T(\mathcal{A}(n))), \dots)$ is an injective semi-conjugacy $\Phi \circ T_{\mathcal{A}} = \sigma \circ \Phi$ from $\{n \in \mathcal{A}\} \cap \mathbb{Z}_{>0}$: $\{\text{orbit infinite}\}$ into X .

(b) 2-adic homeomorphism. Φ extends to a homeomorphism $\hat{\Phi}: \hat{\mathcal{A}} \rightarrow X$ (2-adic topology on $\hat{\mathcal{A}}$, product topology on X).

(c) Density. $\Phi(\mathcal{A} \cap \mathbb{Z}_{>0})$ is dense in X .

Proof. (a) Conjugacy by definition; injectivity: $(k_1, \dots, k_J)$ determines $n \bmod 2^{k_1 + \dots + k_J + J}$ via Lemma A. (b) Surjectivity: admissible word $(k_1, \dots, k_J)$ determines a 2-adic ball $B_J \subset \hat{\mathcal{A}}$; $B_{J+1} \subset B_J$ and $\mathrm{diam}(B_J) \rightarrow 0$; $\mathbb{Z}_2$ complete gives $\bigcap B_J \neq \emptyset$. (c) Each B_J contains infinitely many positive integers (APs). $\square$

Remark. Surjectivity uses compactness in $\mathbb{Z}_2$, *not* in $\mathbb{Z}_{>0}$. Positive integers are a dense arithmetic subset of $\hat{\mathcal{A}}$.

4. Stationary Measure and Entropy

4.1 Theorem 20-H [R]

Theorem 20-H. $\pi(k) = 1/2^j$ is the unique stationary distribution; $E_\pi[k] = 2$, $\mathrm{Var}_\pi[k] = 2$.

Proof. $(\pi P)(j) = \pi(j+1) + \pi(1)/2^j = 2^{-(j+1)} + 2^{-(j+1)} = 2^{-j}$. Uniqueness: irreducible + positive recurrent. $\square$

4.2 Theorem 27 [R]

Theorem 27. $h(\pi, \sigma) = \log 2$.

Proof. $h = \pi(1) \cdot H(\mathrm{Geom}(1/2)) = \frac{1}{2} \cdot \log 2 \cdot E_\pi[k] = \log 2$. $\square$

Note. Topological word entropy is $+\infty$ (infinite alphabet); Theorem 27 is measure-theoretic entropy.

5. The Shift-and-Kill Decomposition

5.1 The Operators

Markov operator P : $(Pf)(j) = f(j-1)$ ($j \geq 2$) and $(Pf)(1) = E_\pi[f]$.

Shift-and-kill S : $(Sf)(j) = f(j-1) \cdot \mathbf{1}_{j \geq 2}$.

The name: S shifts f down by one index and *kills* the value at $j=1$.

5.2 Lemma S1 [R]

Lemma S1. For $f \in L^2(\pi)$ with $E_\pi[f] = 0$: $P^N f = S^N f$ exactly.

Equivalently: $(P^N f)(j) = f(j-N) \cdot \mathbf{1}_{\{j > N\}}$.

Proof by induction on N .

$N=1$: $(Pf)(j \geq 2) = f(j-1) = (Sf)(j)$. $(Pf)(1) = E_\pi[f] = 0 = (Sf)(1)$. $\checkmark$

The key step at $j=1$: the renewal resets to $E_\pi[f] = 0$ the killed value.

Step: $E_\pi[P^N f] = E_\pi[f] = 0$, so $P^{N+1} f = P(P^N f) = S(P^N f) = S^{N+1} f$. $\square$

6. Exact Mixing

6.1 Theorem 25-I [R]

Theorem 25-I.

(a) $\mathrm{Cov}_\pi(k_0, k_N) = 2 \cdot (1/2)^N$ exactly.

Proof. $g = k-2$, $E_\pi[g] = 0$. By S1: $\mathrm{Cov} = \langle g, S^N g \rangle_\pi = \sum_{j > N} (j-2)(j-N-2)/2^j = 2(1/2)^N$. $\square$

[Verified $N=1, \dots, 7$, exact match.]

(b) $|\mathrm{Cov}_\pi(f(k_0), g(k_N))| \leq \|f\|_\infty \|g\|_\infty \cdot (1/2)^N$.

Proof. $|\langle f, S^N g \rangle_\pi| \leq \|f\|_\infty \|g\|_\infty \sum_{j > N} \pi(j) = (1/2)^N$. $\square$

(c) $|P^N f|_{L^2(\pi)} = (1/\sqrt{2})^N \|f\|_{L^2(\pi)}$ for $E_\pi[f] = 0$.

Proof. $\|S^N f\|_\pi^2 = \sum_{j > N} \pi(j) |f(j-N)|^2 = (1/2)^N \|f\|_\pi^2$. $\square$

Remark. L^2 spectral gap $= 1 - 1/\sqrt{2}$; L^∞ mixing rate $(1/2)^N$; exact covariance for k decays at the L^∞ rate. These three statements are all correct and mutually consistent.

7. Exact Information Memory

7.1 Setup

$I_\pi(k_0; k_N) = D_\pi(\mathrm{KL})(P_{\{k_0, k_N\}} | P_{\{k_0\}} \otimes P_{\{k_N\}})$, the mutual information at lag N .

7.2 Theorem IT2a [R]

Theorem IT2a. $\sum_{N=1}^{\infty} I_{\pi(k_0;k_N)} = 2\log 2$ (a finite, explicitly computable value; numerically equal to $h(\pi, \sigma)$).

Proof. From Lemma S1: for $j > N$, $k_N = k_0 - N$ deterministically, giving joint mass $\pi(j)$ at $(j, j - N)$; for $j \leq N$, the renewal makes k_0, k_N independent. Thus:
 $I_{\pi(k_0;k_N)} = \sum_{j > N} \frac{1}{2^j} \log \frac{1}{2^j} \pi(j) \pi(j - N) + \sum_{j \leq N} \frac{(j - N)}{\log 2} \frac{1}{2^j} = 2\log 2 \cdot \left(\frac{1}{2}\right)^N$.
Summing: $\sum_{N=1}^{\infty} I_N = 2\log 2 \cdot \sum_{N=1}^{\infty} (1/2)^N = 2\log 2$. $\square$

[Verified $N=1, \dots, 29$, exact match.]

Correction. An earlier version claimed $\sum I \leq 2/3$ via Pinsker. Pinsker’s inequality $D_{\mathrm{KL}}(P|Q) \geq \frac{1}{2} \|P - Q\|_{\mathrm{TV}}^2$ provides a *lower* bound on KL divergence. The correct value, computed directly above, is $2\log 2 \approx 1.386$ nats.

8. Discussion

9.1 Summary

Result	Content	Method
Theorem EC	$\hat{\Phi} : \hat{\mathcal{A}} \rightarrow X$; dense injection	2-adic compactness
Theorem 20-H	$\pi(k) = 2^{-k}$ unique	Stationarity + ergodicity
Lemma S1	$P^N f = S^N f$ on $\ker E_{\pi}$	1-step induction
Theorem 25-I	$\mathrm{Cov} = 2(1/2)^N$; L^{∞} , L^2	From S1
Theorem 27	$h(\pi) = \log 2$	Markov entropy formula
Theorem IT2a	$\sum I = 2\log 2$ (explicit)	From S1 + KL computation

9.2 Scope and Context

The results are self-contained within the renewal-shift framework. The companion paper (Kamanoi 2026b) examines what a hypothetically divergent orbit would require structurally

— in particular, how divergence would necessitate escaping the finite-memory regime established here.

9.3 Generality

The shift-and-kill identity $P^N = S^N$ holds for any geometric renewal chain. The mutual information formula generalizes: $I_N = -E_{\pi}[\log \pi(k-N)] \cdot p^N$, $\quad \sum_N I_N = \frac{-E_{\pi}[\log \pi(k)] \cdot p}{1-p}$, for general geometric parameter p . The Collatz value $p=1/2$ (from Lemma B) is natural: it gives a symmetric renewal and the clean formula $\sum I = 2 \log 2$.

Concluding Remark

The present work does not resolve the Collatz conjecture. Rather, it isolates and exactly computes the finite-memory structure of the renewal system that underlies it. The remaining obstruction — whether an individual arithmetic orbit can sustain an infinite-memory deviation from the renewal law — appears not purely dynamical, but fundamentally arithmetic.

The present framework therefore suggests that the unresolved core of Collatz is not the average drift, but the possibility of orbitwise arithmetic coherence persisting indefinitely against an exact renewal background.

References

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All numerical verifications performed in Python. Code available upon request.